

From Static Snapshots to Dynamic Symphonies: The Evolution of Agentic and Physics-Guided Generative Paradigms in Protein Design

The field of protein design is undergoing a paradigmatic shift from static, decoupled sequence-structure workflows toward dynamic, joint co-generation frameworks. Historically limited by the "generate-then-fix" mentality, modern approaches now leverage SE(3)-equivariant diffusion models to simultaneously optimize backbone geometry and amino acid sequences, ensuring physical validity and designability [14]. This evolution extends beyond static folds to incorporate conformational dynamics and mechanical properties, treating proteins as flexible ensembles rather than rigid snapshots [11]. Crucially, the integration of physics-informed objectives, such as energy landscape alignment, mitigates structural hallucinations and grounds generative outputs in thermodynamic reality [272]. Furthermore, the frontier is expanding from passive generation to autonomous, agentic systems capable of iterative reasoning and multi-objective optimization. Frameworks like AgentPLM and ProtoCycle now interleave generation with biophysical feedback loops, enabling the precise steering of designs toward complex functional specifications without extensive retraining [135]. Collectively, these advancements signal a transition toward universal biomolecular simulators that unify sequence, structure, dynamics, and function within a single, controllable generative process.

Introduction to Generative Paradigms in Protein Design

The field of protein design has undergone a profound paradigm shift, transitioning from traditional rational design and labor-intensive directed evolution toward data-driven generative artificial intelligence. Historically, protein engineering relied on modifying existing natural proteins or assembling fragments based on physical principles, approaches often constrained by the vastness of sequence space and the prohibitive costs of experimental validation [74]. The emergence of deep learning has fundamentally altered this landscape, enabling the *de novo* creation of biomolecules with specific functions and structures that do not exist in nature [5]. This transition is characterized by the ability of generative models to learn complex distributions from massive protein databases, allowing researchers to navigate the combinatorial explosion of possible sequences and structures with unprecedented efficiency [50]. The impact of this shift is evident in state-of-the-art design protocols now achieving experimental success rates nearing 20%, a significant improvement over previous computational protocols that struggled with low success rates for computationally designed enzymes [24].

Central to this revolution are diverse architectural innovations, particularly diffusion models and protein language models (PLMs), which have surpassed traditional fragment-based and bioinformatics methods. Diffusion models, such as RFDiffusion and Genie, have demonstrated remarkable success in generating novel protein backbones by iteratively denoising random structures into stable, designable folds [3]. These models often leverage SE(3)-equivariant neural networks to ensure that generated structures respect the physical symmetries of three-dimensional space, addressing previous limitations where models produced non-physical geometries like left-handed helices [14]. Concurrently, PLMs, trained on hundreds of millions of evolutionary sequences, have proven adept at capturing the statistical rules of protein evolution, enabling the generation of functional sequences conditioned on taxonomic or functional tags [63]. The integration of these approaches has led to a new era where generative AI serves not just as a tool for creating static structures, but as a comprehensive simulator for engineering the dynamic language of life [77].

A critical evolution in generative paradigms is the move from unimodal generation to joint sequence-structure co-generation. Early methods often treated structure prediction and sequence design as separate steps, typically generating a backbone first and then using inverse folding models to populate it with amino acids [18]. However, this decoupled approach can fail to capture the intricate interdependence between sequence and fold, leading to designs that are either non-foldable or lack the desired function. Recent advancements address this by modeling both modalities simultaneously. For instance, models like CodeFP and ESM3 decode sequence and structure tokens jointly, ensuring that functionality and foldability are realized in tandem [10]. Similarly, frameworks like Fold2Seq utilize joint embeddings to map the complex relationship between 1D sequences and 3D topological folds, improving diversity and reliability compared to structure-only baselines [112]. This co-generative capability is essential for designing proteins with complex dynamic properties or specific binding interfaces, where local sequence changes can have global structural consequences [2].

Despite these advances, significant challenges remain in controlling generative outputs to meet precise user specifications. While generative models excel at sampling from the distribution of natural proteins, they often struggle to explore low-probability regions of the fitness landscape that encode novel, high-value properties [238]. To address this, new strategies for "steering" or guiding models have emerged. Methods such as ProteinGuide allow for the plug-and-play conditioning of pre-trained models on auxiliary properties like stability or enzymatic activity without requiring retraining, effectively blending predictive models with generative priors [43]. Furthermore, agentic frameworks and reinforcement learning approaches are being developed to iteratively refine designs based on feedback loops, mimicking the cycle of directed evolution but entirely *in silico* [197]. These systems enable the optimization of multi-objective criteria, such as binding affinity and solubility, pushing the boundaries of what can be achieved through purely generative means [154].

The scope of generative design is also expanding beyond static monomeric structures to include dynamic ensembles, large assemblies, and complex biomolecular interactions. Traditional design often assumed a rigid backbone, neglecting the conformational flexibility essential for many biological functions [90]. Newer models are beginning to incorporate dynamics directly into the design process, such as VibeGen, which conditions generation on normal mode vibrations to create proteins with tailored mechanical properties [11]. Additionally, the field is witnessing the rise of "world models" capable of designing all-atom interactions between proteins, nucleic acids, and small molecules within a unified

framework, as seen in ODesign and AlphaFold 3 extensions [146]. These developments suggest a future where generative AI serves not just as a tool for creating static structures, but as a comprehensive simulator for engineering the dynamic language of life [13].

The Evolution from Single-Modality to Co-Generation

Historically, the computational design of proteins was defined by a decoupled, sequential workflow that treated structure and sequence as distinct modalities to be optimized in isolation. This traditional paradigm typically involved generating a protein backbone based on geometric constraints or topological specifications, followed by a separate inverse folding step to assign an amino acid sequence capable of stabilizing that structure. While methods such as those utilizing score-based generative models for binding peptide backbones demonstrated the ability to generate novel structures, they often relied on external inverse folding models to produce viable sequences, creating a dependency that could introduce mismatches between the designed backbone and the final sequence [104]. Similarly, approaches like Fold2Seq focused on designing sequences for desired folds but learned sequence and fold embeddings separately, leading to heterogeneous latent domains that struggled to fully capture the complex, bidirectional relationship between one-dimensional sequences and three-dimensional structures [112]. This separation often resulted in designs where the generated sequence failed to fold into the intended backbone or lacked the necessary functional properties, as the sequence design phase was blind to the specific geometric nuances generated in the structural phase. The limitations of this single-modality approach are further highlighted by the observation that standard metrics for evaluating generated backbones, such as ground-truth RMSD, often fail to correlate with physicochemical violations in designed backbones, necessitating a more integrated evaluation framework that considers the interdependence of form and function [104].

The limitations of this single-modality approach have driven a significant shift toward co-generation frameworks, where sequence and structure are modeled jointly within a unified probabilistic distribution. This evolution addresses the fundamental reality that protein sequence and structure are strongly correlated and mutually constraining; a change in one necessitates a corresponding adaptation in the other to maintain stability and function. Recent advancements have introduced models that simultaneously decode sequence and structure tokens, thereby avoiding the pitfalls of non-folding or misfolding inherent in decoupled pipelines. For instance, GeoPro proposes a method to jointly design protein backbone structure and sequence, powered by an equivariant encoder for 3D structure and a sequence decoder guided by geometry, which has successfully discovered novel metalloproteins not present in existing databases [147]. By treating the design problem as a joint optimization task, these models ensure that the generated candidates satisfy both global conformational constraints and local chemical environments simultaneously. Similarly, NAEPro facilitates the joint design of sequence and structure based on automatically detected functional and conserved sites, resulting in proteins with superior binding affinity scores compared to baselines that treat these modalities independently [202].

Further refining this co-generative paradigm, newer architectures have moved beyond simple joint decoding to incorporate multi-modal representations that include surface chemistry, side-chain atomistics, and functional motifs. The model DS-ProGen integrates both backbone geometry and surface-level representations into a deep language model, demonstrating that capturing chemical and geometric constraints simultaneously leads to higher sequence recovery rates and better functional retention compared to methods relying on backbone coordinates alone [100]. Similarly, the emergence of fully atomistic co-design frameworks like Protei?na-Atomi?stica highlights the critical importance of aligned sequence-structure data; by training on synthetic datasets where sequences and structures are consistently paired, these models achieve dramatic improvements in structural diversity and co-designability, effectively removing the need for post-hoc sequence redesign steps that were standard in earlier workflows [177]. This represents a departure from the "generate then fix" mentality of single-modality approaches, moving instead toward a holistic generation process where atomistic details are intrinsic to the model's output. The transition to co-generation also enables more sophisticated conditioning on functional requirements, allowing for the design of proteins with tailored dynamics and binding properties that single-modality models struggle to achieve. Models such as ApexGen simultaneously design peptide binder sequences and structures for target proteins, coupling sequence and structure design at every step of the integration process to ensure tight fitting and high predicted binding affinity [78].

Even in the realm of language models, which traditionally operated on sequences, there is a push toward unified modalities. FoldToken learns a discrete protein language that represents both sequence and structure as unified symbols, enabling GPT-style models to perform co-generation with promising results [124]. Furthermore, CodeFP utilizes a co-generative protein language model that simultaneously decodes sequence and structure tokens, overcoming the suboptimal translation of flat encodings and achieving superior functional consistency and foldability [10]. The integration of dynamics into this co-generative framework is exemplified by VibeGen, which employs an agentic

dual-model architecture to design proteins conditioned on normal mode vibrations, establishing a direct link between sequence and vibrational behavior [11]. Despite these advances, the field continues to grapple with the complexity of modeling the full sequence-structure-function landscape. While joint models have shown that simultaneous decoding improves functional consistency and foldability, challenges remain in scaling these approaches to large assemblies and ensuring robust generalization beyond the training distribution [18]. The move from single-modality to co-generation marks a maturation of the field, shifting from treating proteins as static backbones to be populated with sequences, to viewing them as dynamic, integrated systems where form and function emerge together from a shared generative process. This paradigm shift is essential for realizing the potential of de novo protein design in creating novel therapeutics and enzymes that transcend the limitations of natural evolution.

The Role of Equivariance and Physical Constraints

A defining characteristic of modern generative paradigms in protein design is the rigorous enforcement of physical symmetries and chemical constraints, a necessity that distinguishes viable biological structures from geometrically plausible but physically impossible artifacts. The fundamental challenge lies in navigating the non-Euclidean nature of rotational space while ensuring that generated atoms adhere to strict bond lengths, angles, and thermodynamic stability. Early generative attempts often struggled with these constraints, but recent advancements have established that operating directly on the Lie group $SE(3)$ or ensuring $SE(3)$ invariance is critical for generating viable backbones. This shift represents a move from post-hoc filtering to intrinsic architectural guarantees, where the generative process itself respects the laws of physics. The $SE(3)$ group, consisting of all rotations, reflections, and translations, provides the mathematical foundation for maintaining the physical stability of the $N-C_{\alpha}-C$ frame of each amino acid, ensuring that the relative geometries dictating biochemical function are preserved regardless of the protein's orientation in space [4].

The consensus on the necessity of equivariance is strong, particularly for maintaining the relative geometries that dictate biochemical function. [44] addresses the lack of a principled framework for diffusion on $SE(3)$ by developing FrameDiff, which learns an $SE(3)$ equivariant score over multiple frames to generate monomer backbones up to 500 amino acids without relying on pretrained structure prediction networks. Similarly, [14] introduces Genie, a model that performs discrete-time diffusion using a cloud of oriented reference frames. Genie demonstrates that retaining angular information via $SE(3)$ -equivariant reasoning is essential for capturing chemical chirality and avoiding nonphysical structures like left-handed alpha helices, a common failure mode in models that only enforce reflection invariance. These approaches argue that the explicit modeling of frames is not merely an optimization choice but a physical requirement for capturing the nuanced dependencies of protein folding.

However, a notable contradiction exists regarding the implementation of these symmetries, specifically between explicit equivariant layers and latent space simplifications. While the aforementioned models advocate for explicit geometric reasoning, [69] proposes mapping backbone coordinates directly into a continuous latent space using a non-equivariant Diffusion Transformer. This approach argues that simplifying the optimization pipeline by bypassing explicit equivariance can still achieve state-of-the-art reconstruction quality. Yet, empirical evidence suggests limitations to this simplification. [104] demonstrates through controlled comparisons that reasoning over rotations (frames) is essential for generating diverse antibody CDR loop structures, whereas coordinate-only models fail to capture the necessary structural diversity despite similar RMSD metrics. This suggests that while latent methods offer computational efficiency, explicit $SE(3)$ equivariance remains crucial for functional design, particularly in irregular regions lacking secondary structure.

Beyond global symmetries, the integration of local geometric and thermodynamic constraints has become paramount. Standard Cartesian diffusion often violates bond constraints, prompting a shift toward internal coordinates. [76] shifts the generative process to torsion angles (ϕ, ψ, ω) , guaranteeing perfect local geometry with bond lengths at exactly $3.8 \text{ \AA}$ and employing a refinement step to adjust the Radius of Gyration without violating bond constraints. Furthermore, the noising process itself is being re-evaluated to align with physical principles. [184] argues that standard noising dynamics lack physical grounding, proposing instead a physically motivated non-linear noising process that unfolds proteins into secondary structures while preserving topological integrity and preventing atomic collisions.

The scope of physical constraints also extends to the energy landscape, addressing the issue of "structural hallucinations" where sequences appear valid but are thermodynamically unstable. [272] highlights that language models often produce sequences with high linguistic likelihood but unstable conformations. It proposes a physics-informed alignment framework that grounds optimization in the energy landscape, effectively mitigating hallucinations by recovering biophysical interactions such as hydrophobic core packing. This integration of energy-based objectives ensures that generated structures are not only geometrically correct but also foldable.

Finally, specific applications require tailored constraints to handle complex topologies. [118] enables control over substructure location and orientation by conditioning generation on 3D ellipsoids, allowing for the design of proteins with specific helix fractions. In multi-domain assembly, [191] employs an $SE(3)$ diffusion model to flexibly assemble

individually generated domains, addressing the rugged interdomain interaction landscape. For tasks requiring strict adherence to functional sites, [261] allows motifs to float rigidly during diffusion, automating the design of relative motif positions. Additionally, [142] integrates proximal feasibility updates with ADMM decomposition to ensure strict adherence to bonding and geometric constraints, achieving perfect satisfaction of hard constraints in motif scaffolding tasks. These developments illustrate that the future of generative protein design lies in hybrid architectures that combine the expressivity of equivariant deep learning with the rigor of physical and thermodynamic laws.

Architectural Foundations of Diffusion and Flow Models

The architectural landscape of de novo protein design has undergone a paradigm shift, moving from traditional fragment-based assembly and early Generative Adversarial Networks (GANs) toward sophisticated diffusion and flow-based generative models. This transition is driven by the superior ability of diffusion models to capture complex, high-dimensional distributions of protein structures and sequences, offering unprecedented sample quality and diversity [3]. Foundational to this success is the mathematical rigor of Denoising Diffusion Probabilistic Models (DDPM) and Score-based Generative Models (SGM), which provide a robust framework for iteratively refining noise into biologically plausible conformations [4]. Unlike GANs, which often struggle with mode collapse, diffusion models excel at exploring the vast combinatorial space of protein folds, effectively modeling the "generative modeling trilemma" of quality, diversity, and sampling efficiency [14]. Recently, flow matching has also emerged as a powerful and efficient alternative to traditional diffusion, offering straighter trajectories and faster sampling speeds for biological applications [267].

A critical architectural challenge in protein generation is ensuring physical validity through geometric equivariance. Proteins exist in three-dimensional space where their functional properties are invariant to rotations and translations, necessitating models that respect the $E(3)$ or $SE(3)$ symmetry groups. Early approaches utilizing standard Euclidean networks often failed to maintain the physical stability of the $N-C_{\alpha}-C$ frame, leading to non-physical artifacts such as left-handed α helices [4]. To address this, architectures like Genie and FrameDiff employ $SE(3)$ -equivariant denoisers that operate on clouds of oriented reference frames rather than simple point clouds, thereby preserving angular information and chemical chirality [14][44]. Furthermore, FoldingDiff introduced a representation based on internal torsion angles to inherently satisfy rotational invariance, although this approach sometimes sacrifices global structural coherence compared to Cartesian frame-based methods [53]. Recent advancements have also explored latent space approaches, such as PROTEINAE and SaDiT, which compress structural information into compact latent representations using tokenization or autoencoders, allowing for efficient generation while bypassing the need for explicit equivariant layers during the diffusion process itself [69][66]. Hierarchical approaches like LSD further refine this by separating coarse-grained topology from fine-grained atomic details, enabling better control over large-scale folds [31].

While structure generation has seen remarkable progress, the architectural treatment of protein sequences presents distinct challenges due to their discrete, categorical nature. Traditional continuous diffusion models, dominant in image generation, require adaptation for amino acid sequences. Approaches such as discrete diffusion and masked language modeling have emerged as viable solutions, with models like DPLM and EvoDiff demonstrating that discrete diffusion can effectively learn evolutionary constraints and generate novel, functional sequences [9][46]. However, a tension exists between discrete and continuous representations. Some researchers argue that continuous latent spaces derived from protein language models (pLMs) offer superior reconstruction accuracy and smoother optimization landscapes. For instance, DiMA operates on continuous embeddings from pLMs, leveraging the rich semantic information captured by transformers to facilitate both sequence and structure generation [6]. Similarly, LaGDif bridges the discrete-continuous divide by performing diffusion in a continuous latent graph space before decoding to discrete sequences, achieving state-of-the-art results in inverse folding [215]. Conversely, other works emphasize the biological realism of discrete edit operations, proposing frameworks like DPLM-Evo that explicitly model substitution, insertion, and deletion events to better mimic natural evolutionary trajectories [46].

Beyond the choice of representation, the mechanism of guidance and conditioning remains a pivotal architectural component for practical design tasks. Unconditional generation is insufficient for engineering proteins with specific functions; thus, architectures must support conditional sampling. Classifier-guided diffusion and classifier-free guidance have been adapted from computer vision to steer generation toward desired properties, such as binding affinity or solubility [72][48]. More advanced frameworks utilize context-guided diffusion (CGD) to improve out-of-distribution generalization, enabling the design of proteins that lie beyond the training distribution while maintaining structural integrity [180]. Large-scale models like ProteinAI and ProteinAI-Atomistic employ multi-stage generation processes and unified flow-based frameworks to jointly model sequences and atomistic structures, allowing for better control over large-scale folds and local geometry [173][177].

Despite these architectural strides, contradictions and limitations persist. While equivariant models ensure physical plausibility, they often incur significant computational costs, prompting a trade-off between geometric rigor and sampling efficiency [66]. Additionally, the reliance on synthetic datasets, such as those derived from AlphaFold, introduces potential biases; recent studies highlight that inconsistent sequence-structure pairs in training data can severely degrade the co-designability of generated proteins, necessitating the creation of aligned, consistency-distilled datasets [177]. The field is also grappling with the challenge of modeling conformational dynamics rather than static states. While most diffusion models target single static structures, emerging architectures like EigenFold and AlphaFolding aim to capture conformational ensembles and dynamic trajectories, utilizing harmonic oscillator formulations or 4D diffusion to represent the flexible nature of biological function [107][128]. These developments suggest that future architectural innovations will increasingly focus on multimodal integration, dynamic modeling, and the efficient navigation of heterogeneous fitness landscapes through geometry-aware smoothing mechanisms [133].

Continuous Diffusion and Flow Matching for Backbone Generation

The paradigm of protein backbone generation has shifted decisively from discrete fragment assembly to continuous generative modeling operating directly in high-dimensional geometric spaces. This transition leverages the mathematical robustness of diffusion probabilistic models and flow matching to navigate the complex energy landscapes of protein folding. A central challenge in this domain is ensuring that generated structures adhere to the physical symmetries of Euclidean space, specifically invariance to rotation and translation. To address this, recent architectures have increasingly adopted SE(3)-equivariant neural networks, allowing models to reason about geometric data without bias toward global orientation. For instance, FrameDiff establishes a theoretical framework for diffusion on the SE(3) group, operating on rigid body frames rather than simple point clouds to generate designable monomers up to 500 amino acids without relying on pretrained structure prediction networks [44]. Similarly, Genie utilizes a cloud of oriented reference frames within a discrete-time diffusion process, demonstrating that equivariant reasoning over frames yields structures with superior designability, novelty, and diversity compared to methods relying solely on inter-residue distances or internal coordinates [14]. These approaches contrast with earlier Cartesian methods like ProtDiff, which, despite using equivariant graph neural networks, occasionally produced non-physical structures such as left-handed alpha helices due to reflection-invariant properties inherent in some geometric deep learning implementations [14].

While Cartesian coordinate-based approaches offer direct geometric control, they often struggle with maintaining strict local geometric constraints, such as fixed bond lengths and angles, requiring post-hoc relaxation or complex loss functions. An alternative strategy involves performing diffusion in internal coordinate space, specifically torsion angles, which inherently preserves local geometry. FoldingDiff mirrors the biological folding process by denoising a sequence of torsion angles from a random, unfolded state, thereby alleviating the need for complex equivariant networks while ensuring physically valid bond geometries [53]. This approach stands in contrast to Cartesian methods by trading explicit global equivariance for inherent local validity. To bridge the gap between global shape control and local geometric fidelity, hybrid approaches have emerged. Torsion-Space Diffusion for Protein Backbone Generation with Geometric Refinement proposes a model that predicts torsion angles to guarantee perfect local geometry while employing a separate refinement step to optimize the radius of gyration, achieving 100% bond length accuracy [76]. Furthermore, physically motivated noising processes are being explored to better align generative dynamics with thermodynamic principles; one such method introduces a non-linear noising process grounded in classical physics that unfolds proteins into secondary structures while preserving topological integrity before applying flow matching on SE(3) [184].

Beyond standard diffusion, flow matching has emerged as a computationally efficient alternative for continuous backbone generation, offering faster training and more stable convergence. Flow matching models learn deterministic or stochastic continuous-time dynamics that map a simple source distribution to the complex target distribution of protein structures. FoldFlow exemplifies this approach by utilizing flow matching over 3D rigid motions, offering faster training and more stable convergence compared to traditional diffusion-based approaches while maintaining high sample quality for backbones up to 300 amino acids [264]. The efficiency of flow matching is further enhanced when combined with latent space representations. PROTEINAE introduces a streamlined diffusion autoencoder that maps backbone coordinates into a continuous latent space using a non-equivariant transformer trained with a single flow matching objective, effectively bypassing the computational overhead of explicit equivariance while achieving state-of-the-art reconstruction quality [69]. Similarly, SaDiT accelerates generation by integrating discrete latent structural tokenization with diffusion transformers, utilizing an Invariant Point Attention cache to reuse computed states during iterative sampling [66]. These latent approaches suggest that explicit equivariance in the generative process may not always be necessary if the latent space is sufficiently expressive and the decoding mechanism respects geometric constraints.

The field also recognizes the importance of hierarchical and multiscale modeling to capture the recurring patterns in natural proteins, which standard white-noise diffusion models often overlook. Ophiuchus employs an SO(3)-equivariant coarse-graining model that uses local convolutional coarsening to learn high-level building blocks, subsequently leveraging denoising diffusion in the latent space for efficient sampling [152]. This aligns with the broader observation that natural data exhibit multiscale structures that can be exploited for efficiency. To exploit this, renormalization

group-based diffusion models have been proposed to generate data in a coarse-to-fine manner, significantly accelerating sampling speeds while preserving sample quality by systematically integrating out short-distance modes [237]. Additionally, the integration of physical priors remains critical for ensuring realism; CONFDIFF incorporates a force-guided network with score-based models to ensure generated conformations adhere to equilibrium distributions, addressing the tendency of pure data-driven models to deviate from physical realism [225]. These diverse architectural innovations collectively demonstrate that continuous diffusion and flow matching, when coupled with appropriate geometric representations and physical constraints, provide a powerful foundation for de novo protein backbone design.

Discrete Diffusion for Sequence Modeling

In contrast to continuous spatial diffusion, which operates primarily on Euclidean coordinates for structure generation, discrete diffusion models address the fundamental challenge of generating amino acid sequences by treating the problem as a denoising process over a categorical distribution. This paradigm shift is critical because protein sequences are inherently discrete tokens, and applying Gaussian noise directly to one-hot encoded vectors often results in invalid states or requires complex projection steps. The theoretical underpinnings of these models have been significantly advanced by unifying disparate approaches; specifically, recent work demonstrates that discrete diffusion, Gaussian diffusion, and simplex diffusion can all be viewed as parameterizations of the Wright-Fisher population genetics model [217]. This unification not only clarifies the mathematical relationships between these methods but also enables the training of single models capable of operating across different domains, thereby stabilizing processes that were previously numerically unstable, such as simplicial diffusion, and relieving practitioners of the need to balance domain-specific trade-offs.

The application of discrete diffusion to protein sequence modeling has evolved from simple masking strategies to sophisticated frameworks that mimic biological evolution. Traditional Protein Language Models (pLMs) relying on masked language modeling often suffer from limited generative capacity due to low noise corruption rates during training. To overcome this, the Diffusion Sequence Model (DSM) introduces a masked forward diffusion process inspired by large language diffusion frameworks, enabling robust denoising even at 90% token corruption [7]. This approach allows for the generation of diverse, biomimetic sequences that align with natural amino acid compositions and secondary structures while maintaining high-quality representations for downstream tasks. Furthermore, the concept of "evolutionary" discrete diffusion has been formalized in DPLM-Evo, which explicitly models substitution, insertion, and deletion operations rather than relying solely on masking [46]. By decoupling the latent alignment space from the observed sequence space, DPLM-Evo facilitates variable-length generation and adaptive scaffold growth, addressing a key limitation in earlier models that contradicted the biological intuition of accumulated edits. Similarly, SCISOR utilizes a discrete diffusion process trained to reverse random insertions, effectively learning to shrink proteins while maintaining function, a capability distinct from standard deletion models [52].

Specific architectural innovations have further refined the efficacy of discrete diffusion in sequence space, particularly for inverse folding where the goal is to recover sequences from a fixed backbone. Graph-based discrete diffusion models have achieved state-of-the-art results by incorporating amino acid replacement matrices into the forward process, encoding biologically meaningful prior knowledge of amino acids from their spatial and sequential neighbors [64]. This integration reduces the sampling space and improves sequence recovery compared to discriminative models that struggle to encapsulate the diverse range of plausible solutions. To bridge the discrete-continuous divide, latent space approaches like LaGDif encode protein graphs into a continuous latent space for diffusion before decoding back to discrete sequences, achieving significant improvements in sequence recovery rates by leveraging smooth transitions inherent to continuous spaces [215]. Other works, such as DiMA, operate diffusion on continuous protein language model embeddings, showing that latent continuous diffusion can generalize across various encoders while maintaining high sequence quality [6]. Additionally, the Shortlisting Model (SLM) operates on simplex centroids to reduce generation complexity, demonstrating competitive performance in DNA and protein design by progressively pruning candidate sets [198].

Beyond unconditional generation, discrete diffusion models have proven highly effective for conditional design and optimization tasks, though steering these models requires overcoming the non-differentiable nature of discrete trajectories. Methods such as DRAKES utilize the Gumbel-Softmax trick to enable direct backpropagation of rewards through diffusion trajectories, allowing for the optimization of protein stability and DNA enhancer activity while preserving naturalness [87]. Alternatively, inference-time guidance strategies like DISCRETE DENOISING POSTERIOR PREDICTION (DDPP) cast steering as a probabilistic inference problem, enabling the alignment of protein language models to generate diverse secondary structures without simulation [175]. For multi-objective optimization, frameworks like MP2D integrate constrained Monte Carlo Tree Search with discrete diffusion to navigate conflicting property landscapes, such as balancing antimicrobial activity with low toxicity [60]. Additionally, MCTD-ME combines masked diffusion with tree search guided by multiple experts to enable multi-token planning and efficient exploration in large sequence spaces [130]. These advancements collectively highlight that discrete diffusion is

not merely an alternative to continuous methods but a necessary framework for capturing the combinatorial and categorical nature of biological sequences, offering a unified path toward autonomous protein engineering.

Noise Schedules, Sampling Strategies, and Efficiency

The efficacy of diffusion models in protein design is fundamentally governed by the interplay between noise scheduling, representation geometry, and sampling algorithms, which collectively determine the trade-off between computational tractability and structural fidelity. While standard image-based diffusion frameworks typically rely on linear or cosine noise schedules, emerging evidence suggests these are sub-optimal for the high-dimensional, sparse, and heterogeneous landscapes characteristic of biomolecular data. Research indicates that the sequential and discrete nature of protein representations necessitates tailored noise corruption processes to avoid inefficient training dynamics where reconstruction loss remains trivially low for extended periods [6]. To address this, alternative scheduling mechanisms, such as tangent-based schedules, have been proposed to ensure a more linear increase in reconstruction loss over diffusion time, thereby optimizing the learning signal across all timesteps [6]. Furthermore, the assumption of homogeneous data distribution inherent in fixed global noise levels fails to capture the isolated "islands" of functional sequences within the vast combinatorial protein space. Geometry-aware smoothing techniques, such as Density-Dependent Smoothing, adapt the noise scale to local data density, preventing the oversmoothing of dense functional clusters while facilitating exploration in sparse regions [133]. This data-dependent approach is critical for navigating the sparse evolutionary fitness landscape, where fixed global smoothing assumptions fundamentally limit generative modeling capabilities [133].

Beyond noise scheduling, the efficiency of generation is significantly enhanced by shifting from atomistic Cartesian coordinates to latent, internal, or fractional coordinate representations. Operating in continuous latent spaces allows models to bypass the computational bottlenecks of equivariant networks and complex manifold constraints. For instance, latent diffusion models utilizing protein language model encodings have demonstrated the ability to generate diverse, high-quality sequences with architectures that are agnostic to the specific encoder, effectively decoupling representation learning from the generative process [6]. Similarly, frameworks like PROTEINAE map backbone coordinates directly into compact latent spaces, enabling efficient structure generation without explicit equivariance constraints while maintaining state-of-the-art reconstruction quality [69]. Hierarchical approaches further refine this efficiency by separating coarse and fine-grained details; models such as LSD sample latents to define contact maps before generating atomic coordinates, allowing for explicit control over designability and novelty [31]. In the realm of side-chain packing, torsion-space diffusion models like DiffPack operate directly on torsional angles, guaranteeing perfect local geometry and reducing parameter counts significantly compared to Cartesian baselines [158]. Complementary work in torsion-space diffusion for backbone generation reinforces this by demonstrating that predicting torsion angles inherently preserves bond lengths and angles, eliminating the need for post-hoc geometric refinement [76]. Additionally, discrete latent tokenization strategies, such as those employed by SaDiT, accelerate backbone generation by integrating structural tokenization with diffusion transformers, significantly reducing computational complexity while maintaining theoretical SE(3) equivalence [66]. Further advancing the mathematical foundation of these dynamics, ProT-GFDM employs fractional stochastic processes with superdiffusive properties to improve the capture of long-range dependencies in protein structures, outperforming conventional score-based models in density and coverage [103].

Sampling strategies have also evolved to incorporate physical priors and iterative refinement mechanisms to improve sample quality and adherence to functional constraints. Standard denoising processes are increasingly augmented with force-guided sampling, where physics-based force fields provide energy-based feedback to steer the generation toward low-energy, stable conformations. This approach has proven effective in antibody design, where integrating force field information helps mold complementarity-determining regions to target interfaces despite limited training data [293] [225]. Additionally, inference-time control frameworks leveraging Feynman-Kac steering enable the optimization of non-differentiable objectives, such as binding affinity, by guiding the sampling trajectory without retraining the underlying model [140]. Iterative refinement strategies, such as Reward-Guided Iterative Refinement, introduce a noising and denoising loop at test time to progressively correct errors and optimize downstream reward functions, demonstrating superior performance in protein and DNA design tasks [260].

The integration of planning algorithms with diffusion models represents another frontier in enhancing sampling efficiency and strategic decision-making. Monte Carlo Tree Diffusion with Multiple Experts (MCTD-ME) combines masked diffusion with tree search, utilizing an ensemble of expert models to guide exploration and exploitation in the

vast sequence space. This method addresses the limitations of autoregressive planners by enabling multi-token planning and focusing computational resources on low-confidence regions via pLDDT-guided masking [130]. Furthermore, renormalization group-based diffusion models offer a theoretical framework for accelerating sampling by leveraging multiscale data structures. By reversing renormalization group flows, these models generate samples in a coarse-to-fine manner, significantly reducing the number of iteration steps required compared to conventional denoising diffusion probabilistic models while maintaining or improving sample quality [237]. Context-guided diffusion methods also improve out-of-distribution generalization by leveraging unlabeled data and smoothness constraints, proving effective across continuous, discrete, and graph-structured processes in protein design [180]. Finally, adaptive order policies for masked diffusion learn optimal unmasking sequences, outperforming heuristic ordering in tasks sensitive to token dependency, thereby enhancing the coherence of generated protein sequences [265].

Integration of Language Models and Multimodal Representations

The advent of Protein Language Models (pLMs) has fundamentally shifted computational biology from a reliance on scarce structural annotations to the exploitation of vast, unlabeled sequence databases. By treating amino acid sequences as a linguistic modality, these models capture deep evolutionary constraints and structural propensities inherent in the protein universe [63]. Early iterations primarily focused on unsupervised representation learning, adapting techniques from natural language processing to classify protein properties or predict fitness without explicit structural labels [295]. However, a significant divergence emerged between models optimized for understanding-typically bidirectional auto-encoders-and those designed for generation, which often rely on autoregressive architectures. This fragmentation has spurred the development of unified frameworks capable of handling both modalities simultaneously, such as xTrimoPGLM, which integrates bidirectional attention for understanding with autoregressive objectives for generation at a scale of 100 billion parameters [196].

A critical challenge in the field is the integration of multimodal data, specifically the fusion of sequence, structure, and textual functional descriptions. While sequence-only models like ProGen2 have demonstrated state-of-the-art performance in generating viable sequences by conditioning on taxonomic and functional tags, they often lack the fine-grained structural awareness required for precise *de novo* design [108]. To bridge this gap, researchers have introduced hybrid architectures that incorporate structural information directly into the language modeling framework. For instance, HD-Prot proposes a hybrid diffusion model that embeds continuous-valued structure latents atop a discrete pLM, avoiding the information loss associated with discretizing 3D coordinates into tokens [23]. Similarly, ProtTeX tokenizes sequences, structures, and text into a unified discrete space, enabling joint training via next-token prediction and facilitating structure-in-context reasoning [204]. These approaches contrast with earlier methods that treated structure as a separate input or relied on post-hoc filtering, instead promoting a seamless co-design of sequence and structure [42].

The capacity for controllable generation represents another frontier where multimodal integration is essential. Standard pLMs tend to sample from dense regions of the training distribution, limiting their ability to explore rare, high-value sequences with specific desired properties [125]. To address this, reinforcement learning (RL) frameworks such as DPO_pLM have been adapted to align pLM outputs with external oracles, optimizing for properties like binding affinity or thermostability without requiring additional training data [125]. Furthermore, the integration of natural language instructions has enabled more intuitive control mechanisms. Models like InstructProtein and ProtDAT leverage instruction tuning and cross-attention mechanisms to generate proteins based on free-text descriptions, effectively aligning human language with protein semantics [183][203]. This capability allows for the design of ligand-binding proteins or enzymes with specific functions described purely in natural language, a task that was previously intractable for sequence-only models [148].

Despite these advances, tensions remain regarding the optimal architecture for multimodal integration. Diffusion-based models, such as DPLM-2 and DiMA, argue that discrete diffusion processes offer a more principled approach to joint sequence-structure modeling compared to autoregressive methods, providing superior performance in inverse folding and motif scaffolding tasks [25][6]. Conversely, other studies suggest that smaller, efficient models fine-tuned with Low-Rank Adaptors (LoRA) can match the performance of massive foundation models while significantly reducing computational costs, challenging the necessity of billion-parameter scales for specific design tasks [83]. Additionally, the interpretability of these multimodal systems remains a concern, as the "black box" nature of deep learning models hinders trust in their design decisions. Emerging work in Explainable AI (XAI) seeks to illuminate the internal mechanisms of pLMs, categorizing their roles as evaluators or engineers to ensure that generated sequences adhere to biophysical constraints [153]. The convergence of these diverse methodologies-ranging from continuous diffusion and reinforcement learning to natural language instruction tuning-signals a maturing field where the integration of multimodal representations is not merely an additive feature but a foundational requirement for the next generation of protein engineering tools.

Latent Space Diffusion on Protein Embeddings

A paradigm shift in computational protein design has emerged through the transition from diffusion processes operating on raw sequences or explicit 3D coordinates to those navigating the continuous latent spaces of pre-trained protein language models (pLMs). This approach addresses the combinatorial complexity inherent in discrete amino acid spaces and the computational burden of equivariant 3D diffusion by leveraging the rich semantic and structural encodings captured by pLMs. The DiMA framework exemplifies this innovation, demonstrating that continuous diffusion on protein embeddings enables robust sequence and structure generation across diverse encoder architectures, ranging from 8 million to 3 billion parameters [6]. By operating in a continuous manifold, DiMA facilitates the application of established score-based techniques, such as classifier-free guidance, without the need for discrete approximations, thereby offering a more stable training regime compared to discrete counterparts [6]. This stability is crucial for generalizing across tasks such as motif scaffolding and fold-conditioned design, where the preservation of global context is paramount. Furthermore, recent work on diffusion sequence models confirms that masked diffusion can unify high-quality representation learning with generative capabilities, producing biomimetic sequences that align with expected secondary structures and functions even under high corruption rates [7].

The integration of language models and multimodal representations has further evolved through architectures that jointly model sequence and structure within unified latent frameworks, though strategies for handling structural continuity differ significantly. HD-Prot proposes a hybrid diffusion protein language model that embeds a continuous-valued diffusion head atop a discrete pLM, enabling seamless operation with both discrete sequence tokens and continuous structure latents [23]. This approach circumvents the information loss associated with vector quantization, allowing for high-fidelity joint sequence-structure modeling [23]. In contrast, DPLM-2 extends discrete diffusion protein language models to accommodate both modalities by converting 3D coordinates into discrete tokens via lookup-free quantization, learning the joint distribution of sequence and structure to eliminate the need for two-stage generation approaches [25]. These divergent paths highlight a consensus that capturing the intricate relationships between sequence and structure requires a unified representation space, yet they reflect an ongoing debate regarding the optimal treatment of continuity. Complementing these efforts, PROTEINAE streamlines structure encoding by mapping backbone coordinates directly to a compact latent space using a non-equivariant Diffusion Transformer, achieving state-of-the-art reconstruction quality and enabling efficient structure generation that competes with leading structure-based methods [69]. Similarly, SaDiT accelerates protein backbone generation by integrating discrete latent structural tokenization with diffusion transformers, demonstrating that tokenization can significantly reduce generation complexity while maintaining theoretical SE(3) equivalence [66].

Beyond general generation, latent space diffusion has proven particularly effective for specialized tasks such as backmapping coarse-grained models to all-atom resolutions and designing peptide binders, where efficiency and conformational exploration are critical. Latent Diffusion Backmapping (LDB) leverages denoising diffusion within a latent space to reconstruct all-atom structures from coarse-grained representations, bypassing the need for complex equivariant manipulations and facilitating broader exploration of conformational space [236]. Similarly, BackDiff utilizes conditional score-based diffusion with geometric representations to achieve generalization across various coarse-grained models without retraining [250]. In the realm of peptide design, PepEDiff introduces a zero-shot framework that generates binder sequences directly in a continuous latent space derived from pretrained embeddings, conditioning on receptor information without relying on intermediate structure predictions [110]. This structure-free approach enhances diversity and avoids the cascading errors often associated with misaligned structural intermediates, demonstrating the versatility of latent diffusion in bridging small molecule and protein design [213]. Additionally, hierarchical approaches like LSD separate coarse and fine-grained details by sampling latents decoded into contact maps before generating atomic coordinates, allowing for explicit balancing of designability and diversity [31].

Theoretical advancements continue to refine how latent spaces are navigated and optimized, particularly in addressing the heterogeneity of biological sequence landscapes. While standard diffusion assumes homogeneous noise schedules, recent work suggests that biological sequence spaces are heterogeneous, requiring geometry-aware smoothing to avoid oversmoothing functional clusters or fragmenting sparse regions [133]. Although primarily discussed in discrete contexts, these insights into the geometry of the latent manifold inform the design of continuous diffusion schedulers and guidance mechanisms. Additionally, the superposition of diffusion models allows for the combination of multiple

pre-trained models at the inference stage within the latent space, enabling logical operations like AND and OR to guide generation toward specific properties without retraining [231]. This capability is crucial for multi-objective optimization, where balancing conflicting properties such as stability and binding affinity is essential. However, challenges remain in aligning continuous latent dynamics with the discrete nature of biological function, as the decoding step can introduce discretization errors. Methods like Planner Aware Path Learning (PAPL) address the mismatch between training and inference paths in discrete diffusion, a concern that translates to latent spaces where the decoding step introduces discretization errors [119]. The convergence of these methods indicates a robust trend toward leveraging the abstract, continuous representations of pLMs to unlock generative capabilities that are both computationally efficient and biologically coherent [7].

Unified Sequence-Structure Tokenization

The integration of language models with multimodal protein representations necessitates a fundamental rethinking of how structural data is encoded alongside amino acid sequences. To achieve true multimodal understanding, recent advancements have moved beyond treating sequence and structure as separate modalities, instead proposing unified tokenization schemes that either discretize 3D coordinates into a shared vocabulary or embed continuous structure latents directly within the language modeling framework. A primary challenge in this domain is the inherent conflict between the discrete nature of language modeling and the continuous, geometric reality of protein structures. Early approaches often relied on converting 3D coordinates into discrete tokens to fit standard transformer architectures, yet this process frequently incurs a substantial loss of fidelity regarding fine-grained structural details and correlations [188]. Models such as DPLM-2 address this by utilizing lookup-free quantization-based tokenizers to convert 3D coordinates into discrete tokens, enabling the joint learning of sequence and structure distributions through discrete diffusion [25]. Similarly, Yeti employs a compact tokenizer based on lookup-free quantization trained with a flow matching objective, demonstrating that high codebook utilization and token diversity can be achieved with significantly fewer parameters than previous baselines, thereby facilitating scalable multi-modal training [216].

However, the discretization of continuous structural space remains a point of contention, with arguments suggesting that vector quantization inevitably limits the performance potential of multimodal protein language models by discarding high-fidelity information. In response, alternative frameworks like HD-Prot propose a hybrid approach that incorporates continuous structure tokens, or high-fidelity protein structure latents, avoiding vector quantization entirely [23]. HD-Prot embeds a continuous-valued diffusion head atop a discrete protein language model, allowing for the seamless operation of both discrete sequence tokens and continuous structure tokens within a unified absorbing diffusion process. This duality is further explored in Flow Autoencoders Are Effective Protein Tokenizers, which introduces Kanzi, a flow-based tokenizer that replaces complex, bespoke components with a single flow matching loss, stabilizing training and outperforming existing tokenizers on reconstruction metrics while simplifying the architecture to use standard attention mechanisms [138]. The debate between discrete and continuous representations is also central to Diffusion on Language Model Encodings for Protein Sequence Generation, which argues that continuous diffusion on protein language model representations offers superior reconstruction accuracy and stability compared to discrete alternatives, providing a universal framework that generalizes across diverse encoder architectures [6]. Furthermore, PROTEINAE demonstrates that mapping protein backbone coordinates directly into a continuous, compact latent space via a non-equivariant Diffusion Transformer can achieve state-of-the-art reconstruction quality, bypassing the need for explicit equivariance and discrete tokenization entirely [69].

Beyond the binary choice of discrete versus continuous tokens, recent research emphasizes the importance of the tokenization strategy's scope and adaptability. Traditional local neighborhood pooling methods are increasingly being replaced by global tokenization strategies. Adaptive Protein Tokenization presents a method where successive tokens contribute increasing levels of detail to a global representation, effectively mitigating error accumulation and enabling task-specific adaptation of information content [30]. This shift towards global and hierarchical representations is echoed in Ophiuchus, which utilizes hierarchical coarse-graining to model sequence-motif interactions efficiently, departing from single-resolution graph modeling [152]. Furthermore, the concept of a unified "protein language" is expanded in FoldToken, which projects both residue types and structures into a discrete space to create "FoldTokens," effectively transforming the sequence-structure pair into a single modality suitable for GPT-style modeling [124]. This unification allows for any-to-any generation capabilities, as seen in HelixProtX, which leverages large multimodal models to transform any input protein modality into any desired output modality, supported by robust tokenization strategies [83].

The efficacy of these unified tokenization schemes is increasingly validated through their application in complex generative tasks such as motif scaffolding, inverse folding, and co-design. For instance, ProtTeX tokenizes sequences, structures, and textual information into a unified discrete space, enabling general large language models to perceive and process protein structures through sequential text input and output [204]. This approach facilitates multimodal reasoning where structural information serves as an intermediate component, significantly improving function prediction accuracy. Conversely, models like SaDiT integrate discrete latent space tokenization with diffusion transformers to accelerate backbone generation, utilizing an IPA Token Cache to optimize invariant point attention layers [66]. The tension between reconstruction fidelity and generative diversity is also addressed by learning the

language of protein structure through vector-quantized autoencoders that achieve high-fidelity reconstructions while enabling the generation of novel, designable structures [115]. Ultimately, the field is converging on the view that the quality of multimodal models is fundamentally upper-bounded by the expressiveness and fidelity of the underlying tokenization scheme, driving continuous innovation in how protein geometry is translated into the language of machine learning [216].

Enhancing Representation with Masked Diffusion and Reasoning

Masked diffusion has emerged as a transformative paradigm in computational biology, effectively bridging the historical divide between high-quality representation learning and robust generative capacity. Traditional Protein Language Models (pLMs), particularly those based on Masked Language Modeling (MLM), have excelled at learning semantic representations from vast unlabeled sequence datasets but have historically suffered from limited generative capabilities due to training objectives restricted to low levels of noise corruption [7]. Conversely, autoregressive models offer strong generation but often fail to capture the global, bidirectional context essential for modeling the complex epistatic interactions found in protein structures [7]. Masked diffusion models address these limitations by unifying these objectives, enabling models to denoise sequences from high corruption rates-up to 90%-while maintaining or exceeding the representational quality of standard MLMs on downstream tasks [7]. This unified framework allows for the generation of diverse, biomimetic sequences that align with expected amino acid compositions and secondary structures, establishing masked diffusion as a natural fit for protein modeling [7]. Furthermore, theoretical work has unified discrete, Gaussian, and simplicial diffusion under the Wright-Fisher population genetics model, providing a stable mathematical foundation for these discrete sequence operations [217].

The efficacy of masked diffusion is significantly enhanced by advancements in reasoning and sampling strategies that move beyond simple random unmasking. Standard diffusion processes often assume a uniform random denoising path, which can be suboptimal for tasks sensitive to token ordering, such as combinatorial optimization or protein design [265]. To mitigate this, recent research has introduced adaptive order policies where a lightweight policy network learns to prioritize unmasking positions where the denoiser is most confident, thereby improving performance on complex biological sequences [265]. However, a critical disconnect often exists between training and inference; while training assumes random paths, inference frequently employs "planners" to select favorable generation trajectories. This mismatch can degrade performance, prompting the development of Planner Aware Path Learning (PAPL), which aligns the training objective with planned inference dynamics, yielding substantial improvements in protein sequence generation metrics [119]. Extending this reasoning capability further, Monte Carlo Tree Diffusion with Multiple Experts (MCTD-ME) integrates diffusion models with tree search algorithms, utilizing an ensemble of expert models to guide the denoising process through complex sequence spaces, effectively balancing exploration and exploitation in protein design tasks [130].

Beyond optimizing the diffusion trajectory itself, the integration of reasoning mechanisms allows for precise steering of generative outputs toward specific biological properties without extensive retraining. Techniques such as Discrete Denoising Posterior Prediction (DDPP) frame the steering problem as probabilistic inference, enabling the alignment of models with non-differentiable reward functions like binding affinity or stability [175]. Similarly, reinforcement learning frameworks like DPO_pLM adapt direct preference optimization to guide pLMs toward rare, high-value regions of the sequence space, such as specific enzymatic activities or nanomolar binders, which unsupervised models typically fail to sample [125]. These methods demonstrate that masked diffusion models can be dynamically adjusted at test time to satisfy multi-objective constraints, a capability further refined by approaches like PROUD, which ensures generated samples reside on the Pareto front of conflicting property objectives while maintaining high generation quality [289]. Additionally, iterative refinement frameworks allow for reward-guided optimization at test time, progressively correcting errors to improve design fidelity [260].

The versatility of masked diffusion extends to multimodal integration, where reasoning over both sequence and structure is required. Models like DPLM-2 and HD-Prot incorporate continuous structure tokens or discrete structural representations alongside sequences, allowing for joint sequence-structure co-generation and inverse folding tasks that leverage the full geometric context of proteins [25; 23]. This multimodal reasoning is crucial for applications such as ligand-conditioned design, where models must simultaneously satisfy geometric constraints imposed by small molecules and sequence constraints imposed by the protein backbone [21]. Additionally, the application of masked diffusion to specialized domains, such as membrane protein design via MeMDLM or antibody design through germline-absorbing diffusion, highlights the framework's adaptability to distinct biological priors and constraints [22; 126]. By combining the representational richness of bidirectional attention with the generative flexibility of iterative denoising and advanced reasoning policies, masked diffusion models provide a comprehensive solution for the challenges of modern protein engineering.

Conditional and Controllable Protein Design Strategies

While unconditional generative models effectively explore the vast protein universe, practical applications in biotechnology and medicine demand precise navigation of sequence space to satisfy specific constraints such as function, binding affinity, taxonomic origin, or structural dynamics. The transition from passive exploration to intent-driven design has catalyzed the development of diverse conditional generation strategies, ranging from tag-based conditioning and adapter-based fine-tuning to inference-time steering and reinforcement learning. Early approaches established the feasibility of controllable generation by training large-scale language models on sequences augmented with metadata. For instance, ProGen introduced a conditional language model trained on 280 million sequences with taxonomic and functional tags, demonstrating that fine-grained control over molecular function and cellular component is achievable through standard conditioning mechanisms [63]. Building on this foundation, TaxDiff incorporated taxonomic information directly into transformer layers within a diffusion framework, achieving superior performance in generating structurally stable proteins specific to biological species compared to unconditional baselines [8]. Similarly, ProCALM utilizes adapters to fine-tune protein language models for conditional generation based on versatile functional representations, including natural language descriptions, allowing for generalization to unseen functions [186].

Beyond simple tagging, recent advancements have focused on integrating structural and geometric constraints to ensure that generated sequences are not only functionally relevant but also foldable. Structure-informed approaches like LM-DESIGN implant lightweight structural adapters into sequence-based models, enabling them to design sequences for given folds with high accuracy by leveraging both evolutionary and structural knowledge [101]. More complex architectural innovations include ProtComposer, which conditions generation on spatial layouts specified by 3D ellipsoids, allowing users to control the location, size, and orientation of protein substructures [118]. In the realm of binder design, where the interaction interface is critical, models like Prot42 and SeedProteo have demonstrated the ability to generate high-affinity binders for specific targets, with SeedProteo achieving picomolar-level affinities in wet-lab assays [82; 39]. Furthermore, specialized frameworks such as DiffPepBuilder and ApexGen address the unique challenges of peptide binder design by co-designing sequences and structures or utilizing SE(3)-equivariant diffusion to stabilize generated peptides with disulfide bonds [73; 78].

A significant divergence in the field exists between methods that require retraining or fine-tuning and those that employ inference-time steering to guide pre-trained models. Steering techniques offer a plug-and-play alternative, avoiding the computational cost of retraining massive foundation models. Activation steering, originally developed for large language models, has been adapted to protein language models to direct outputs toward targeted properties without additional training [251]. Similarly, the Feynman-Kac steering framework extends inference-time control to diffusion models, guiding sampling toward desirable energetic features while maintaining diversity [140]. ProteinGuide provides a unified mathematical framework for guiding various generative models, including masked language models and autoregressive models, by blending predictive models of desired properties with the generative prior, successfully demonstrated in enhancing the activity of adenine base editors [43]. These methods contrast with reinforcement learning approaches like DPO_pLM, which fine-tune model weights to maximize rewards from external oracles, effectively shifting the model's distribution toward rare, high-value regions of the sequence space such as nanomolar binders [125].

The complexity of control objectives has also expanded from single attributes to multi-objective and combinatorial optimization. Frameworks like SAGE-Prot iteratively combine autoregressive generation with scoring models to optimize multiple properties simultaneously, such as binding affinity and thermal stability, leveraging curriculum learning to adapt to increasingly complex design goals [154]. CFP-GEN addresses the limitation of single-condition constraints by integrating multimodal conditions, including functional annotations and structural constraints, to generate multifunctional proteins [12]. Moreover, emerging models are beginning to tackle dynamic properties; VibeGen, for example, conditions generation on normal mode vibrations to design proteins with tailored dynamical behaviors, bridging the gap between static structure and functional motion [11]. Despite these advances, challenges remain in balancing exploration and exploitation, as noted in surveys highlighting that many models still struggle to sample from low-probability regions encoding valuable properties without explicit steering mechanisms [238; 5]. The

integration of explainable AI is also becoming crucial to understand the decision boundaries of these conditional models, ensuring that generated designs are not only optimal but also interpretable and safe [153].

Conclusion

The field of protein design has undergone a profound paradigm shift, transitioning from labor-intensive, rational engineering and decoupled computational workflows to sophisticated, data-driven generative artificial intelligence. This evolution is defined by the move from unimodal, sequential processes-where backbones were generated before sequences were designed-to integrated co-generative frameworks that model sequence, structure, and function as a unified probabilistic distribution. Early approaches often failed to capture the intricate interdependence between amino acid identity and three-dimensional topology, leading to designs that were either non-foldable or functionally inert. In contrast, modern architectures utilize joint tokenization schemes and unified latent spaces to ensure that foldability and functionality emerge simultaneously. By treating proteins as dynamic, integrated systems rather than static scaffolds, these advanced models have significantly improved experimental success rates and expanded the accessible regions of the protein fitness landscape.

Architectural innovations have been central to this progress, yet they reveal critical tensions between geometric rigor and computational efficiency that will define future research. The adoption of SE(3)-equivariant diffusion models and flow matching has resolved fundamental issues regarding physical symmetry, ensuring generated structures adhere to the laws of physics and avoid non-physical artifacts. However, a notable divergence exists between these explicit geometric approaches and latent space methods that bypass equivariance for speed. While explicit frame-based reasoning is essential for capturing the nuances of irregular loops and binding interfaces, latent space models offer scalability and smoother optimization landscapes. Similarly, the field is navigating the trade-off between discrete and continuous representations; discrete diffusion models better mimic the categorical nature of amino acids and evolutionary edits, whereas continuous latent spaces facilitate the application of gradient-based guidance. The resolution of these architectural contradictions-likely through hybrid models that leverage the strengths of both-remains a pivotal area for development.

Beyond architecture, the frontier of protein design has expanded from unconditional generation to precise, multi-objective controllability. The field has moved beyond simple tag-based conditioning to embrace complex steering strategies, including inference-time guidance, reinforcement learning, and retrieval-augmented mechanisms. These approaches allow researchers to navigate low-probability, high-value regions of the sequence space, optimizing for conflicting objectives such as binding affinity, thermostability, and solubility without retraining foundational models. The emergence of agentic frameworks marks a further leap, where large language models act as autonomous planners that iteratively reason, critique, and refine designs based on biophysical feedback. These systems emulate the decision-making loops of human engineers, facilitating the design of molecules with tailored mechanical properties and allosteric behaviors that were previously inaccessible to traditional evolutionary or computational methods.

Looking forward, the scope of design is shifting from static structure prediction to the engineering of dynamic ensembles and complex biomolecular interactions. The next generation of "world models" promises to unify the design of proteins, nucleic acids, and small molecules within a single, coherent simulation environment capable of predicting conformational trajectories and thermodynamic ensembles. However, significant challenges remain in bridging the gap between computational metrics and experimental reality. Current evaluation standards often rely on geometric proxies that frequently fail to correlate with functional success or physicochemical validity. The development of robust, holistic benchmarks that integrate dynamics, experimental validation, and standardized datasets is essential to validate these models and ensure their generalizability. Ultimately, the convergence of hybrid physics-AI approaches, multimodal integration, and autonomous agentic reasoning signals a new era where generative AI serves not merely as a design tool, but as a comprehensive simulator for engineering the dynamic language of life.